

HAR Kaggle Competition — Model Run Summary

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Overview

This document summarises every model run submitted to the HAR Kaggle competition (macro F1 metric, 6-class activity recognition). Each run was evaluated with **Leave-One-User-Out (LOUO) cross-validation** on 60 training users (5 folds, 12 users each, round-robin assignment), and the best models were submitted to Kaggle.

Data: Each sample is a 300-timestep window with 6 channels: `mean_x`, `mean_y`, `mean_z`, `std_x`, `std_y`, `std_z`. Classes: 0 = sit/stand, 1 = walk flat, 2 = walk down, 3 = walk up, 4 = running, 5 = other.

Key tension: The mean channels contain the gravity component, which depends on phone orientation per user. Per-user normalization removes this but also removes the gravity direction signal that distinguishes upstairs from downstairs.

Feature Group Abbreviations

The following shorthand is used in the table below. All statistics (stats9) compute: mean, std, min, max, range, median, IQR, skewness, kurtosis.

Abbreviation	Description	Count
<code>std_ch</code>	stats9 of <code>std_x</code> , <code>std_y</code> , <code>std_z</code>	27
<code>mag_std</code>	stats9 of $ \text{std} = \sqrt{(\text{sx}^2 + \text{sy}^2 + \text{sz}^2)}$	9
<code>mag_mean</code>	stats9 of $ \text{mean} = \sqrt{(\text{mx}^2 + \text{my}^2 + \text{mz}^2)}$	9
<code>jerk_ch</code>	stats9 of <code>jerk_x,y,z</code> (=diff of mean channels)	27
<code>mag_jerk</code>	stats9 of $ \text{jerk} $	9
<code>seg_mag</code>	temporal segments (10+20) of <code>mag_std</code> , <code>mag_jerk</code>	120
<code>seg_std_ch</code>	10 segments (mean+std each) of <code>sx</code> , <code>sy</code> , <code>sz</code>	60
<code>seg_jerk_ch</code>	10 segments of <code>jx</code> , <code>jy</code> , <code>jz</code>	60
<code>ac_jerk</code>	autocorr of <code>mag_jerk</code> at 7 lags (1,2,5,10,20,30,60)	7
<code>ac_std_ch</code>	autocorr of <code>sx</code> , <code>sy</code> , <code>sz</code> at 4 lags each	12
<code>spectral</code>	FFT features (dom freq, entropy, low/mid/high power) for <code>mag_jerk</code> , <code>mag_std</code> , <code>sx</code> , <code>sy</code> , <code>sz</code>	25
<code>xcorr</code>	cross-correlations between jerk pairs and std pairs	6
<code>zerocross</code>	zero-crossing rate of <code>mag_jerk</code>	1
<code>peaks</code>	peak rate of <code>mag_jerk</code>	1
<code>user_ctx</code>	z-score deviation of window stats vs user's own mean	45
<code>sma</code>	signal magnitude area (L1 norm) for mean, std, jerk channels	27
<code>jerk_asym</code>	mean of positive / negative values of <code>jx</code> , <code>jy</code> , <code>jz</code>	6

Abbreviation	Description	Count
trend	first_60_mean, last_60_mean, last-first per channel (6 ch)	18
axis_stat	stats18 + diff8 per axis channel (mean_x/y/z, std_x/y/z)	156
acc_mag	stats18 + diff8 for acc_mag, std_mag	52
cross_mag	stats18 + diff8 for xy/xz/yz_mag and std variants	156
sum_feat	stats18 + diff8 for mean_sum, std_sum	52
fft_peak_5ch	FFT (6) + peak (4) features for acc_mag, std_mag, std_x/y/z	50
win_10×5ch	10 windows × 5 stats (mean,std,min,max,energy) for 5 channels	250
tilt	tilt angle stats over window	6
covcorr	pairwise covariance + correlation of mean_x/y/z	6
mean_xyz_fft	FFT features of mean_x, mean_y, mean_z	18
step_rhythm	inter-peak interval stats of std_mag time series	4

The **core 373-feature set** (used run07–run22) = std_ch + mag_std + mag_mean + jerk_ch + mag_jerk + seg_mag + seg_std_ch + seg_jerk_ch + ac_jerk + ac_std_ch + spectral + xcorr + zerocross + peaks = **373**.

Run Summary Table

The table is split into two parts — Part A covers scores, environment and model setup; Part B covers features.

Part A — Scores, Environment, and Model Setup

Run	Kaggle F1	Environment	# Feat	Norm	CV	Augmentation	Ensemble
Baseline A	0.7531	Kaggle	~100	Window	—	No	1 LGB
Baseline B	0.7366	Kaggle	~100	Window	—	No	1 LGB
run03	—	Local→Kaggle ¹	~216	Window	LOUO 5f	No	CNN + 1 LGB
run04	0.7042	Kaggle ²	~216	Window	LOUO 5f	No	CNN + 1 LGB
run05	0.7337	Kaggle ³	~1132	Window ×2	LOUO 5f	No	10 LGB
run06	0.7501	Kaggle ³	373	None	LOUO 5f	No	4 LGB configs
run07	0.7707	Kaggle ³	373	Per-user	LOUO 5f	No	3 LGB configs
run07imp	0.7519	Local	150–200	Per-user	Local	No	15 LGB
run08	0.7559	Kaggle ³	373	Per-user	LOUO 5f	TTA + noise (C2,C4)	LGB
run08(1)	0.7524	Kaggle ³	373	Per-user	Strat. 5f	SMOTE	LGB
run12	0.7295	Kaggle ³	~150	Per-user	LOUO 5f	No	TCN+Transf.+LGB
run15	0.7565	Local (CSV)	50	Per-user	Local	No	15 LGB
run17	0.7400	Kaggle	~428	Per-user	LOUO 5f	No	30 LGB + thresh. opt
run18	0.7738	Kaggle	418	Per-user	LOUO 5f	Noise C2,C4,C5	15 LGB + 9 XGB + thresh. opt
run19	0.7392	Kaggle	~364	Per-user	LOUO 5f	Yes	15 LGB + 9 XGB + thresh. opt
run19_2s	0.5342	Local	373	Per-user	Local	No	2-stage LGB

Run	Kaggle F1	Environment	# Feat	Norm	CV	Augmentation	Ensemble
run20	0.7593	Kaggle	445	Per-user	LOUO 5f	Noise C2,C4,C5	15 LGB + 9 XGB + 5 RF + thresh. opt
run21	—	Local	—	—	—	—	Diagnostic only
run22	0.7691	Kaggle	424	Per-user	LOUO 5f	Noise C2,C4,C5	15 LGB + 9 XGB + thresh. opt
run23	0.7792	Kaggle	382	None	LOUO 5f	Noise C2×5,C4×5,C5×5	15 LGB + 9 XGB + 1 RF + thresh. opt
run24	0.7618	Kaggle	~417	None	LOUO 5f	Noise (same as run23)	15 LGB + 9 XGB + thresh. opt
run25	0.7633	Local+Kaggle	382	None	LOUO 5f	No	5 LGB + 2 XGB + thresh. opt
run26	0.7703	Local+Kaggle	412	None	LOUO 5f	No	5 LGB + 2 XGB + thresh. opt
run27	0.7895	Local+Kaggle	716	None	LOUO 5f	No	5 LGB + 2 XGB + thresh. opt
run28	0.7785	Local+Kaggle	750	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run29	0.7853	Local+Kaggle	750	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run30	0.7893	Local+Kaggle	716	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run31	0.7906	Local+Kaggle	716	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt

Part B — Feature Groups and Key Changes

Run	Feature Groups (see legend)	Key Change / Purpose
Baseline A/B	Basic channel stats	First notebook submissions
run03	ch-stats9, seg(5/10/20), autocorr, slope, xcorr, ZCR, spectral(5×6)	First LOUO-CV; CNN+LGB; CSV loading
run04	Same as run03	Switch to NPZ loading
run05	Same ×2 (raw + window-norm concatenated)	Dropped CNN; massive feature space
run06	core373 (no raw mean channels)	User-invariant features; no normalization
run07	core373	Per-user normalization; first good result
run07imp	core373 → feat. selection (150–200)	Aggressive feature pruning; local only
run08	core373	TTA (5 shifts) + minority oversampling
run08(1)	core373	SMOTE; wrong CV (stratified not LOUO)
run12	core373 → selected (50–200)	TCN + Transformer deep learning attempt
run15	core373 → selected (50)	86.6% feature reduction; CSV loading
run17	core373 + slope(mag), half-compare, 5-seg, 2nd-jerk (+55)	First threshold opt (Nelder-Mead); 30-model LGB
run18	core373 + user_ctx(45) = 418	+User context z-scores; LGB+XGB; augmentation
run19	core373 – mag_mean(9) 364	Ablation: removing mag_mean; no user_ctx
run19_2s	core373	Two-stage C2 cascade; failed (0.53)
run20	core373 + user_ctx(45) + sma(27) = 445	+SMA (L1 magnitude); +RF in ensemble
run21	—	Diagnostic visualization only
run22	core373 + user_ctx(45) + jerk_asym(6) = 424	+Jerk asymmetry (pos/neg per axis)
run23	core373 – mag_mean(9) + trend(18) = 382	Dropped normalization; trend features; strong aug
run24	run23 + extra stats (energy,MAD,RMS,q05,q10,q90,q95) 417	+7 stats/signal; hurt (correlated, noise)
run25	same as run23 = 382	Ablation: no augmentation; fewer models
run26	run23 + seg-energy(sx,sy,sz) = 412	+Segment energy; no augmentation

Run	Feature Groups (see legend)	Key Change / Purpose
run27	axis_stat(156)+acc_mag(52)+cross_mag(156)+cuboid(52)+fft_peak(50)+hyst(150) = 716	Neighborhood + fft
run28	run27 + tilt(6)+covcorr(6)+mean_xyz_fft(18)+step_rhythm(4) = 750	+34 physics feats; stronger reg; +RF; reg kept RF
run29	same as run28 = 750	Reverted regularization; partial recovery
run30	same as run27 = 716	Reverted to 716 feat (34 new redundant); kept RF
run31	same as run27/30 = 716	LR 0.03→0.01; patience 100→300; best result

Kaggle Data Paths

Footnote	Input path detection	Output path
¹ run03	Searches /kaggle/input/*/train/train/ for CSV directories	/kaggle/working/
² run04	Hardcoded:	/kaggle/working/
³ run05–run08	/kaggle/input/train-data/train_data.npz, /kaggle/input/test-data/test_data.npz	
run09–run16	glob("/kaggle/input/**/*.*npz", recursive=True)	/kaggle/working/ if Path("/kaggle/working").exists()
run17–run22	Checks /kaggle/input/, /kaggle/input/train-data/, /kaggle/input/test-data/, /kaggle/input/har-data/	Hardcoded /kaggle/working/
run23–run31	Same search as + local fallback in outputs/	Path("/kaggle/input").exists() → /kaggle/working/; else outputs/ (laptop)

Kaggle Score Progression

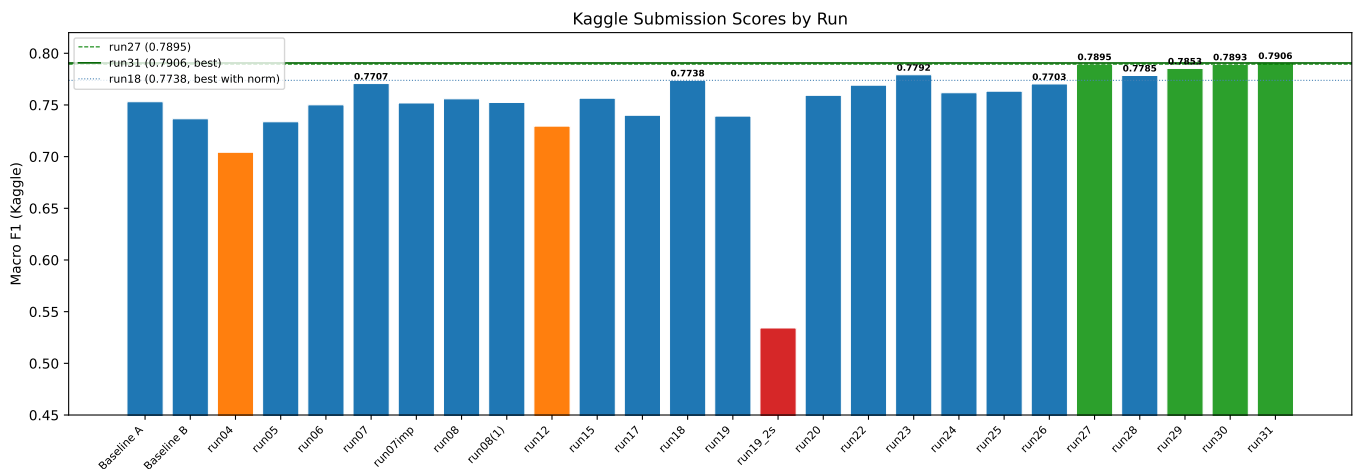


Figure 1: Kaggle macro F1 score per submitted run. Baseline A/B are pre-run notebook submissions.

Key Transitions

Phase 1: Architecture exploration (run03–run08)

Starting from CNN+LGB ensembles loading original CSV files, the early runs explored within-window normalization, feature concatenation (raw + normalized), and TTA/SMOTE augmentation. The CNN was dropped after run04/05 as it consistently hurt Kaggle generalization. The critical discovery was **per-user normalization** (run07, 0.7707): normalizing each user’s windows by that user’s own global mean/std removes the cross-user DC offset while preserving within-user activity differences.

Phase 2: Feature expansion with per-user normalization (run17–run22)

Run17 introduced threshold optimization (Nelder-Mead on OOF probabilities) and a larger 30-model ensemble. Run18 (0.7738) added user-contextual features (+45 z-score deviation features) and a LGB+XGB ensemble with augmentation. Subsequent runs added SMA (run20) and jerk asymmetry (run22) but neither improved substantially, suggesting the 373+45 feature set was near saturation given per-user normalization.

Phase 3: Dropping per-user normalization (run23–run26)

The key insight from the reference notebook (Kaggle score 0.7923) was that **per-user normalization removes the gravity direction**, which is the most discriminative signal for distinguishing walk_up from walk_down. Run23 dropped normalization, removed mag_mean (rotation-invariant, loses gravity direction), and added trend features (first/last 60 timestep mean per channel). Result: 0.7792 (+0.005 vs run18 without augmentation advantage). Removing augmentation (run25: 0.7633) confirmed that **augmentation is essential** when generalizing to unseen test users despite higher CV scores without it.

Phase 4: Notebook-exact feature set (run27–run31)

Run27 implemented the reference notebook’s full 716-feature set (axis-specific mean/std stats+diff, cross-magnitudes, window features, FFT+peaks) achieving 0.7895. Attempts to add 34 new features (tilt, axis covariance, mean FFT, step rhythm) in run28/29 hurt performance — the new features were geometrically redundant with the cross_magnitudes group (0.7785/0.7853). Run30 added a Random Forest to the ensemble (0.7893), and run31 lowered the LGB learning rate (0.03→0.01) with wider early-stopping patience (100→300) to allow more boosting iterations, achieving the best result: **0.7906**.